

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Touch Display
Manufacturer: CleverProducts
Firmware Version: N/A
Model(s): Clevertouch Pro 65", Clevertouch Pro 75", Clevertouch Pro 86"

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.05.0000-b007	2.01.0002-b005	1.2.3

Version History

Module Version	Date	Notes
1_0_0_0	4/28/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example: InterfaceName.connectionCounter = 5
- For Serial Over Ethernet support, SerialOverEthernetClass must be used. Example: InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', PortNum, Model='ModelName')

Control Commands, States, and Qualifiers:

AspectRatio	'16:9'	'4:3'	'Point to point'
AudioMute	'On'	'Off'	
Input	'TV' 'VGA 2' 'HDMI 2' 'Android'	'AV' 'VGA 3' 'HDMI 3' 'HDMI (4K*2K)'	'VGA 1' 'HDMI 1' 'PC'
PCPower	'On'	'Off'	
Power	'On'	'Off'	
RemoteControlFunctions	'WIN' 'Alt+F4' 'NUM_3' 'NUM_6' 'NUM_9' 'Refresh' 'Menu' 'Up' 'Right' 'Back' 'VOL+' 'Page Down' 'F3' 'F6' 'F9' 'F12' 'Yellow'	'Space' 'NUM_1' 'NUM_4' 'NUM_7' 'NUM_0' 'Input' 'Delete' 'Down' 'Enter' 'CH+' 'VOL-' 'F1' 'F4' 'F7' 'F10' 'Red' 'Blue'	'Alt+Tab' 'NUM_2' 'NUM_5' 'NUM_8' 'Display' 'Home' 'Energy' 'Left' 'Point' 'CH-' 'Page Up' 'F2' 'F5' 'F8' 'F11' 'Green'
TVChannelCommand	'0' – '99'		
Volume	0 to 100 in steps of 1		

Status Available:

AudioMute	'On'	'Off'	
ConnectionStatus	'Connected'	'Disconnected'	
Input	'TV'	'AV'	'VGA 1'
	'VGA 2'	'VGA 3'	'HDMI 1'
	'HDMI 2'	'HDMI 3'	'PC'
	'Android'	'HDMI (4K*2K)'	'WHDI'
PCPower	'On'	'Off'	'Sleep'
	'Hibernate'		
Power	'On'	'Off'	
Volume	0 to 100 in steps of 1		

Cable and Adapter Requirements:

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device:

Serial communication:

Port Type: RS-232

Baud Rate: 9600

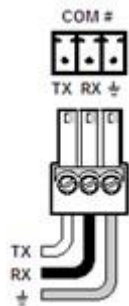
Data Bits: 8

Parity: None

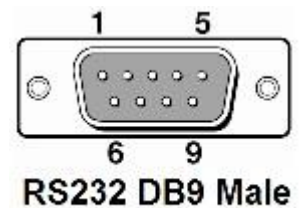
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



General Notes:

Appendix A. Set Commands

Aspect Ratio 16:9	\xAA\xBB\xCC\x08\x00\x00\x08\xDD\xEE\xFF
Aspect Ratio 4:3	\xAA\xBB\xCC\x08\x01\x00\x09\xDD\xEE\xFF
Aspect Ratio Point to point	\xAA\xBB\xCC\x08\x07\x00\x0F\xDD\xEE\xFF
Audio Mute Off	\xAA\xBB\xCC\x03\x01\x01\x05\xDD\xEE\xFF
Audio Mute On	\xAA\xBB\xCC\x03\x01\x00\x04\xDD\xEE\xFF
Input Android	\xAA\xBB\xCC\x02\x0A\x00\x0C\xDD\xEE\xFF
Input AV	\xAA\xBB\xCC\x02\x02\x00\x04\xDD\xEE\xFF
Input HDMI (4K*2K)	\xAA\xBB\xCC\x02\x0D\x00\x0F\xDD\xEE\xFF
Input HDMI 1	\xAA\xBB\xCC\x02\x06\x00\x08\xDD\xEE\xFF
Input HDMI 2	\xAA\xBB\xCC\x02\x07\x00\x09\xDD\xEE\xFF
Input HDMI 3	\xAA\xBB\xCC\x02\x05\x00\x07\xDD\xEE\xFF
Input PC	\xAA\xBB\xCC\x02\x08\x00\x0A\xDD\xEE\xFF
Input TV	\xAA\xBB\xCC\x02\x01\x00\x03\xDD\xEE\xFF
Input VGA 1	\xAA\xBB\xCC\x02\x03\x00\x05\xDD\xEE\xFF
Input VGA 2	\xAA\xBB\xCC\x02\x04\x00\x06\xDD\xEE\xFF
Input VGA 3	\xAA\xBB\xCC\x02\x0B\x00\x0D\xDD\xEE\xFF
PC Power Off	\xAA\xBB\xCC\x09\x00\x00\x09\xDD\xEE\xFF
PC Power On	\xAA\xBB\xCC\x09\x01\x00\x0A\xDD\xEE\xFF
Power Off	\xAA\xBB\xCC\x01\x01\x00\x02\xDD\xEE\xFF
Power On	\xAA\xBB\xCC\x01\x00\x00\x01\xDD\xEE\xFF
Remote Control Functions Alt+F4	\xAA\xBB\xCC\x07\x1F\x00&\xDD\xEE\xFF
Remote Control Functions Alt+Tab	\xAA\xBB\xCC\x07\x1D\x00\$\xDD\xEE\xFF
Remote Control Functions Back	\xAA\xBB\xCC\x07\x0A\x00\x11\xDD\xEE\xFF
Remote Control Functions Blue	\xAA\xBB\xCC\x07_\x00f\xDD\xEE\xFF
Remote Control Functions CH-	\xAA\xBB\xCC\x07\x09\x00\x10\xDD\xEE\xFF
Remote Control Functions CH+	\xAA\xBB\xCC\x07\x02\x00\x09\xDD\xEE\xFF
Remote Control Functions Delete	\xAA\xBB\xCC\x07@\x00G\xDD\xEE\xFF
Remote Control Functions Display	\xAA\xBB\xCC\x07\x1C\x00#\xDD\xEE\xFF
Remote Control Functions Down	\xAA\xBB\xCC\x07M\x00T\xDD\xEE\xFF
Remote Control Functions Energy	\xAA\xBB\xCC\x07N\x00U\xDD\xEE\xFF
Remote Control Functions Enter	\xAA\xBB\xCC\x07J\x00Q\xDD\xEE\xFF
Remote Control Functions F1	\xAA\xBB\xCC\x07E\x00L\xDD\xEE\xFF
Remote Control Functions F10	\xAA\xBB\xCC\x07Y\x00`\xDD\xEE\xFF
Remote Control Functions F11	\xAA\xBB\xCC\x07W\x00^\xDD\xEE\xFF
Remote Control Functions F12	\xAA\xBB\xCC\x07\x08\x00\x0F\xDD\xEE\xFF
Remote Control Functions F2	\xAA\xBB\xCC\x07\x12\x00\x19\xDD\xEE\xFF
Remote Control Functions F3	\xAA\xBB\xCC\x07Q\x00X\xDD\xEE\xFF
Remote Control Functions F4	\xAA\xBB\xCC\x07[\x00b\xDD\xEE\xFF
Remote Control Functions F5	\xAA\xBB\xCC\x07D\x00K\xDD\xEE\xFF
Remote Control Functions F6	\xAA\xBB\xCC\x07P\x00W\xDD\xEE\xFF
Remote Control Functions F7	\xAA\xBB\xCC\x07C\x00J\xDD\xEE\xFF
Remote Control Functions F8	\xAA\xBB\xCC\x07\x1A\x00!\xDD\xEE\xFF
Remote Control Functions F9	\xAA\xBB\xCC\x07\x04\x00\x0B\xDD\xEE\xFF
Remote Control Functions Green	\xAA\xBB\xCC\x07]\x00d\xDD\xEE\xFF
Remote Control Functions Home	\xAA\xBB\xCC\x07H\x00O\xDD\xEE\xFF

Remote Control Functions Input	\xAA\xBB\xCC\x07\x07\x00\x0E\xDD\xEE\xFF
Remote Control Functions Left	\xAA\xBB\xCC\x07\x00P\xDD\xEE\xFF
Remote Control Functions Menu	\xAA\xBB\xCC\x07\x0D\x00\x14\xDD\xEE\xFF
Remote Control Functions NUM_0	\xAA\xBB\xCC\x07\x1B\x00"\xDD\xEE\xFF
Remote Control Functions NUM_1	\xAA\xBB\xCC\x07\x00\x00\x07\xDD\xEE\xFF
Remote Control Functions NUM_2	\xAA\xBB\xCC\x07\x10\x00\x17\xDD\xEE\xFF
Remote Control Functions NUM_3	\xAA\xBB\xCC\x07\x11\x00\x18\xDD\xEE\xFF
Remote Control Functions NUM_4	\xAA\xBB\xCC\x07\x13\x00\x1A\xDD\xEE\xFF
Remote Control Functions NUM_5	\xAA\xBB\xCC\x07\x14\x00\x1B\xDD\xEE\xFF
Remote Control Functions NUM_6	\xAA\xBB\xCC\x07\x15\x00\x1C\xDD\xEE\xFF
Remote Control Functions NUM_7	\xAA\xBB\xCC\x07\x17\x00\x1E\xDD\xEE\xFF
Remote Control Functions NUM_8	\xAA\xBB\xCC\x07\x18\x00\x1F\xDD\xEE\xFF
Remote Control Functions NUM_9	\xAA\xBB\xCC\x07\x19\x00 \xDD\xEE\xFF
Remote Control Functions Page Down	\xAA\xBB\xCC\x07\x0F\x00\x16\xDD\xEE\xFF
Remote Control Functions Page Up	\xAA\xBB\xCC\x07B\x00I\xDD\xEE\xFF
Remote Control Functions Point	\xAA\xBB\xCC\x07\x06\x00\x0D\xDD\xEE\xFF
Remote Control Functions Red	\xAA\xBB\xCC\x07\x00c\xDD\xEE\xFF
Remote Control Functions Refresh	\xAA\xBB\xCC\x07L\x00S\xDD\xEE\xFF
Remote Control Functions Right	\xAA\xBB\xCC\x07K\x00R\xDD\xEE\xFF
Remote Control Functions Space	\xAA\xBB\xCC\x07F\x00M\xDD\xEE\xFF
Remote Control Functions Up	\xAA\xBB\xCC\x07G\x00N\xDD\xEE\xFF
Remote Control Functions VOL-	\xAA\xBB\xCC\x07A\x00H\xDD\xEE\xFF
Remote Control Functions VOL+	\xAA\xBB\xCC\x07\x03\x00\x0A\xDD\xEE\xFF
Remote Control Functions WIN	\xAA\xBB\xCC\x07\x0B\x00\x12\xDD\xEE\xFF
Remote Control Functions Yellow	\xAA\xBB\xCC\x07^\x00e\xDD\xEE\xFF
TV Channel Command 99	\xAA\xBB\xCC\x05\x00ch\xDD\xEE\xFF
Volume 0	\xAA\xBB\xCC\x03\x00\x00\x03\xDD\xEE\xFF
Volume 100	\xAA\xBB\xCC\x03\x00dg\xDD\xEE\xFF

Appendix B. Query Commands

Audio Mute	\xAA\xBB\xCC\x03\x03\x00\x06\xDD\xEE\xFF
Input	\xAA\xBB\xCC\x02\x00\x00\x02\xDD\xEE\xFF
PC Power	\xAA\xBB\xCC\x09\x02\x00\x0B\xDD\xEE\xFF
Power	\xAA\xBB\xCC\x01\x02\x00\x03\xDD\xEE\xFF
Volume	\xAA\xBB\xCC\x03\x02\x00\x05\xDD\xEE\xFF